

# Networking Basics — Exam & Presentation Prep Notes

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Written so you can **understand first, present second**. Read top to bottom once tonight; each section ends with a line you can say out loud tomorrow.

How to use this: the 🗣️ line is your "say it in the room" sentence. The rest is the understanding behind it. If you only remember the 🗣️ lines, you will still sound like you know the topic — because you will.

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## 1. What a network actually is

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Forget the textbook line for a second. A network is just **two or more devices that agreed on rules for talking to each other**.

- The "devices" can be anything: your phone and earbuds, a laptop and a printer, or millions of servers across continents.

- The "rules" are called **protocols**. A protocol is like a language plus

etiquette: who speaks first, how you say "I didn't hear you, repeat that", how you say goodbye. Without protocols, a printer from one company and a laptop from another would just stare at each other.

- The connection itself can be **cables** (fast, stable), **radio waves**

(Wi-Fi, convenient, shared airspace), or both chained together.

The single most useful sentence in networking:

🗣️ "A network doesn't need the internet. Your laptop, a printer, and a home router talking to each other — that's already a network. The internet is just the biggest one."

**Why the deck starts here:** everything later (IP addresses, DNS, routers) is just machinery that makes this simple idea work at planetary scale.

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## 2. Why networks exist — the four reasons

People don't build networks because they love cables. Four motivations, in the order your audience actually cares about:

1. **Share expensive things.** One printer, one internet connection, one file server — used by many people. This is the oldest reason and still the most convincing example: "one community photocopier instead of one per desk."
1. **Communicate.** Email, chat, calls, video meetings. Every message you sent today rode a network.
1. **Use services on other people's computers.** Websites, maps, streaming, cloud apps — your device is a window; the real computer is far away.
1. **Manage everything centrally.** One admin updates fifty office PCs from one desk instead of walking to each one.

 "Ask yourself what you did on your phone this morning — every single one of those things needed a network."

## 3. Types of networks — it's just about size

PAN → LAN → WLAN → MAN → WAN. The names sound intimidating; the idea is trivially simple: **how big is the area?**

| Type | Full name                 | Think of it as...                 | Real example                                         |
|------|---------------------------|-----------------------------------|------------------------------------------------------|
| PAN  | Personal Area Network     | Your personal bubble, arm's reach | Phone ↔ smartwatch, phone ↔ earbuds                  |
| LAN  | Local Area Network        | One building                      | Home Wi-Fi, office network                           |
| WLAN | Wireless LAN              | The same LAN, minus cables        | Home Wi-Fi over radio                                |
| MAN  | Metropolitan Area Network | One city or campus                | University linking its buildings                     |
| WAN  | Wide Area Network         | Countries and continents          | ISP backbone; <b>the internet is the largest WAN</b> |

The trap question is "what's the difference between LAN and WLAN?" — answer: **only the medium**. Same size, same job; one uses cables, the other uses radio. A WLAN *is* a LAN.

💡 "PAN is your pockets, LAN is your building, MAN is your city, WAN is the world. WLAN is just a LAN without cables."

## 4. The cast of characters — network devices

Learn the **jobs**, not the boxes — a home router stuffs several jobs into one plastic case, but the jobs stay distinct.

| Device              | Works at   | Job in one line                                           | Analogy                                |
|---------------------|------------|-----------------------------------------------------------|----------------------------------------|
| <b>Hub</b>          | Layer 1    | Shouts every message out of every port                    | Town crier yelling in the square       |
| <b>Switch</b>       | Layer 2    | Learns which device is on which port, delivers only there | Receptionist who knows every room      |
| <b>Router</b>       | Layer 3    | Connects <i>different</i> networks, picks the best path   | Post office sorting mail by city       |
| <b>Access Point</b> | Layer 1-2  | Lets wireless devices join the wired network              | Translator between radio and cable     |
| <b>Firewall</b>     | Layers 3-7 | Blocks traffic that breaks the rules                      | Security guard checking IDs            |
| <b>Modem</b>        | Layer 1-2  | Translates the ISP line into digital for your router      | Interpreter at a border crossing       |
| <b>Repeater</b>     | Layer 1    | Re-amplifies a dying signal                               | Relay runner carrying the baton onward |

Two confusions that cost marks:

- **Hub vs switch:** a hub *broadcasts* — every device hears everything

(noisy, insecure, slow). A switch *learns* — it builds a table of "this MAC address lives on port 3" and sends each frame only there.

- **Switch vs router:** a switch connects devices **inside one network**

(same street). A router connects **different networks together** (different cities).



"A hub shouts to everyone, a switch delivers to the right room, a router carries mail between cities."

## How a switch learns (favourite viva question)

1. A frame arrives from device A on port 1 → the switch writes down

"A lives on port 1" in its **MAC table**.

1. The destination is unknown → the switch **floods** the frame out of every port except the one it came in on.

1. The real destination replies → the switch learns *\*its\** port too.

2. Next time, no flooding — direct delivery. Entries **age out** after a few minutes so moved devices don't get lost.

## 5. LAN, MAN, WAN and how they connect

- **LAN** is yours: one building, you own everything, fast and free.
- **WAN** belongs to telecom companies: they spent billions laying fibre

across countries, and you **rent** a slice of it. Your ISP is literally a company selling you a doorway onto their WAN.

- Between them sits the **router** — the device whose entire job is standing on the border of two networks and forwarding between them.



"Your LAN is your house. A WAN is the highway system. Your router is your front door — and your ISP owns the highway."

## 6. Topologies — who is wired to whom

Topology = the **shape** of connections.

| Topology   | Shape                        | Verdict                         |
|------------|------------------------------|---------------------------------|
| <b>Bus</b> | One shared cable, all tap in | Dead — one cut kills everything |

| Topology      | Shape                                    | Verdict                                                            |
|---------------|------------------------------------------|--------------------------------------------------------------------|
| <b>Star</b>   | Everything plugs into one central switch | ✅ What everyone builds: one failure hurts one device               |
| <b>Ring</b>   | Circle, each device relays onward        | Rare — one break breaks the ring                                   |
| <b>Mesh</b>   | Everyone connects to everyone            | Only where downtime is unthinkable; needs $n \cdot (n-1)/2$ cables |
| <b>Hybrid</b> | Mix, e.g. star of stars                  | ✅ Real campuses and enterprises                                    |

Why star won: failure is **isolated** (one cable dies, one PC goes dark, the rest never notice), and adding a device means one cable, not rewiring the building. Mesh math impresses examiners: 10 devices fully meshed need 45 cables — that's *\*why\** nobody fully meshes.

💡 "Bus is one road everyone shares — one crash blocks all. Star is every house with its own driveway to one roundabout. That's why every office uses star."

## 7. Transmission media — the actual wires (and air)

| Medium                              | Feel                                       | Weakness                                        |
|-------------------------------------|--------------------------------------------|-------------------------------------------------|
| <b>Twisted pair</b><br>(Cat5e/6/6a) | Up to 10 Gbps over ~100 m                  | Distance-limited                                |
| <b>Coaxial</b>                      | Decent, noise-resistant                    | Bulky, mostly legacy                            |
| <b>Fibre optic</b>                  | Enormous speed × distance, immune to noise | Pricey ends, hates sharp bends                  |
| <b>Radio / Wi-Fi</b>                | Convenient, mobile                         | Shared airspace, interference-prone             |
| <b>Satellite</b>                    | Reaches anywhere                           | Latency — light needs ~0.5 s for the round trip |
| <b>Bluetooth / NFC</b>              | Tiny data, tiny power                      | Metres of range by design                       |

**Guided vs unguided** is just "signal confined in something" (cables) vs "signal let loose" (air and space).

💡 "Copper is a country road, fibre is a bullet train made of light, Wi-Fi is shouting across a crowded room, satellite is shouting at the sky and waiting half a second for the echo."

## 8. The OSI model — the 7-layer cake 🍰

Hardest-looking, easiest-mark slide. The trick: **each layer only talks to the layer directly above and below it**, and each layer wraps the parcel a little more.

Mnemonic, bottom-up:

**Please Do Not Throw Sausage Pizza Away → Physical, Data Link, Network, Transport, Session, Presentation, Application.**

| # | Layer        | Does what                                 | Examples                                   |
|---|--------------|-------------------------------------------|--------------------------------------------|
| 7 | Application  | What the *user* sees                      | HTTP, DNS, browsers                        |
| 6 | Presentation | Translates, encrypts, compresses          | TLS/SSL, JPEG, MP4                         |
| 5 | Session      | Opens and manages conversations           | Logins, video-call sessions                |
| 4 | Transport    | End-to-end delivery, chopped and numbered | <b>TCP</b> (reliable) vs <b>UDP</b> (fast) |
| 3 | Network      | Addressing + best path across networks    | <b>IP, routers</b>                         |
| 2 | Data Link    | Hop-to-hop delivery on one link           | <b>MAC, switches</b>                       |
| 1 | Physical     | Raw bits as electricity, light, radio     | Cables, hubs, Wi-Fi radios                 |

The word examiners want: **encapsulation**. Going down, each layer wraps the previous parcel with its own header (data → segment → packet → frame → bits). Coming up, each layer unwraps its own header. Nested envelopes.

💡 "OSI is a 7-layer cake where every layer only talks to its neighbours. Going down, each layer puts the parcel in a bigger envelope. Coming up, each layer opens its own envelope."

## 9. TCP/IP — the model that actually runs the world

OSI is the textbook; **TCP/IP is the reality**. Four layers, and everything on the internet uses it: **Link → Internet → Transport → Application**.

Mapping to OSI, since exams love this: Link  $\approx$  layers 1+2, Internet = layer 3, Transport = layer 4, Application  $\approx$  layers 5+6+7 merged.

💡 "OSI is the 7-layer classroom diagram. TCP/IP is the 4-layer machine that actually delivers your WhatsApp message."

## 10. MAC addresses — the permanent name tag

- A MAC address is **burned into the network card at the factory**. 48 bits, written as 6 pairs of hex: `3C:22:FB:12:9A:01`.
- First 3 bytes say **who made it** (the manufacturer block), last 3 bytes are that card's **serial number**.
- MACs work **only inside one local network**. Routers strip them off at every hop — a MAC never crosses the internet.

💡 "A MAC address is a factory serial number etched on your phone — permanent, unique, but only useful to the people in this room."

## 11. IP addresses — the postal address of the internet

If MAC is the permanent serial number, IP is the **delivery address** — assigned by the network, changeable, and the thing routers actually read.

- **IPv4**: 32 bits, 4 "octets": `192.168.1.10`, each 0–255. Total space: ~4.3 billion — which the world **ran out of**, hence NAT and IPv6.
- **IPv6**: 128 bits in hex ( `2001:db8::1` ). Space so vast every grain of sand could have addresses. `::` compresses the zeros.
- **Classes**: A = `1–126` (huge), B = `128–191` (medium), C = `192–223` (small, 254 hosts — your home LAN). D = multicast, E = experimental. `127.x.x.x` is **loopback** — "myself" ( `127.0.0.1` = localhost).

- **Private ranges (memorise these three):** `10.x.x.x`, `172.16–31.x.x`,

`192.168.x.x`. Never appear on the public internet — they're the "extension numbers" used inside buildings.

💡 "MAC is the serial number etched at the factory; IP is the desk you get assigned when you walk into the office — it changes when you move."

## 12. Subnetting — dividing one parking lot into sections

A subnet mask tells you **which part of the IP is the street name and which part is the house number**. `/24` means "first 24 bits are network, last 8 are hosts" — same as writing `255.255.255.0`.

- `/24` → 256 addresses, 254 usable (first = network ID, last = broadcast).

That's your home network.

- `/16` → 65,536 addresses (a university). `/8` → 16.7 million.
- **CIDR** (`/24` style) replaced classes because classes were wasteful — why

give a company needing 300 addresses a whole Class B of 65,000?

Exam formula: usable hosts =  $2^{(\text{host bits})} - 2$ . For `/26`:  $32 - 26 = 6$  host bits →  $2^6 - 2 = \mathbf{62 \text{ usable}}$ .

💡 "Subnetting is splitting one big parking lot into marked sections so one fender-bender doesn't block the whole lot."

## 13. TCP vs UDP — certified mail vs postcard

|           | TCP (certified mail  ) | UDP (postcard)                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Guarantee | Delivered, in order, no duplicates                                                                        | Best effort — may vanish or reorder                              |
| How       | Handshake + sequence numbers + ACKs + retransmission                                                      | Fire and forget                                                  |
| Speed     | Slower (all that checking)                                                                                | Fast (no checking)                                               |
| Used by   | Web, email, files — where loss is unacceptable                                                            | Video calls, gaming, streams, DNS — where speed beats perfection |

The famous **3-way handshake**: SYN → SYN-ACK → ACK. ("Can we talk?" "Yes, can you?" "Yes.") Only *\*then\** does data flow. Tearing down uses FIN.

Why video calls use UDP: losing 1 frame in 60 is invisible; *\*waiting\** for a retransmitted frame freezes the whole call. For email, one lost byte corrupts everything — so TCP.

💡 "TCP is certified mail with tracking and signature. UDP is a postcard — faster, cheaper, and if it gets lost, nobody sends a search party."

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## 14. Protocols and ports — languages and doors

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A protocol is an **agreed way of speaking at one layer**. Greatest hits: **HTTP/HTTPS** (web), **FTP** (files), **SMTP** (sending mail), **POP3/IMAP** (receiving mail), **DNS** (names → addresses), **DHCP** (auto-addressing).

**Ports** are door numbers on a machine: HTTP = 80, HTTPS = 443, FTP = 21, SSH = 22, DNS = 53, SMTP = 25. **IP finds the building; the port finds the right door.** Ranges: 0-1023 = well-known standards, 1024-49151 = registered, 49152-65535 = temporary per-connection picks.

💡 "Protocols are layered languages. The port number is which door to knock on."

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## 15. DNS — the phonebook of the internet 📞

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You type `google.com`; computers need `142.250.x.x`. **DNS translates names to numbers**, and the lookup is a chain:

Your cache → hosts file → ISP resolver → root ("who handles .com?") → TLD ("who handles google.com?") → authoritative server ("it's 142.250.x.x").

Each level caches the answer, so the full chain runs rarely. `nslookup` and `dig` let you watch it happen.

💡 "DNS is directory assistance: you say a name, it gives you the number — and it remembers answers so it doesn't ask twice."

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## 16. DHCP — the hotel front desk 🏨

Your phone joins Wi-Fi owning nothing — no address, no gateway, no DNS. **DHCP hands it everything** in four steps, **DORA**: Discover ("any hotel here?") → Offer ("room 192.168.1.20 free") → Request ("I'll take it") → ACK ("confirmed — here's your key and checkout time"). The "checkout time" is the **lease**: addresses are borrowed, recycled when devices leave.

💡 "DHCP is a hotel front desk: your phone walks in with no room, and four messages later it has a key, a checkout time, and directions to the lobby."

## 17. Opening a website — the grand tour 🌐

The slide that ties the \*entire deck\* together. Walk it slowly — it's the best 3 minutes of your presentation. Opening `https://example.com`:

1. **DNS** turns the name into an IP.
2. Your PC checks: same network? (mask comparison). No → send to the

**gateway router**; the **switch** delivers the frame there by **MAC** (ARP resolved the gateway's MAC earlier).

1. The router **NATs** your private IP to its public one and forwards across

ISP routers, each picking the best path by IP.

1. **TCP handshakes** with the server (SYN → SYN-ACK → ACK), then **TLS**

encrypts everything (the 🔒 and the S in HTTPS).

1. **HTTP** requests the page; the server answers with the files. The

**firewall** waved port 443 through at each boundary.

Every earlier slide appears here: MAC, IP, switch, router, DNS, ports, TCP, NAT, firewall. Say that out loud — audiences love the callback.

💡 "Opening one website uses everything we learned: DNS finds it, MAC crosses the room, IP crosses the world, TCP guarantees it, TLS locks it, and the firewall waves port 443 through."

## 18. Firewall, NAT, security, troubleshooting — the closing act

**Firewall = the security guard with a guest list.** Packet filtering (Layer 3) blocks by IP/port — cheap and fast. Stateful inspection (Layer 4) remembers live connections — the standard office firewall. Deep inspection (Layer 7) reads inside the parcel — knows YouTube from malware on the same port.

**NAT = one street address, many apartments.** IPv4 ran out (~4.3 billion addresses, far more devices), so your router stamps its one public IP on every outbound packet and keeps a table to route replies back to the right private device. That's why your laptop is `192.168.x.x` yet reaches the whole internet — and why hosting a server at home needs **port forwarding** (explicit inbound rules).

**Security = layers, not one magic box** (defence in depth): firewall at the border, VPN over hostile networks, IDS watching / IPS blocking inside, VLANs isolating zones so one breach doesn't spread, encryption so stolen data is unreadable. Fence, guard dog, locked rooms, safe — all at once.

**Troubleshooting = elimination, inside → out:** cables/Wi-Fi → got an IP? (`ip addr`) → ping the gateway → ping `8.8.8.8` → resolve names? (`nslookup`) → where does it die? (`traceroute`). The killer split: if numbers work but names don't, it's **always DNS**.

💡 "Security is layers — fence, guard, locked rooms, safe. Troubleshooting is a funnel: your room, your building, the street, the city. The fault is wherever the signal stops."

## 19. Tomorrow's cheat sheet (read this last)

| Asked...          | Answer in one breath                                                     |
|-------------------|--------------------------------------------------------------------------|
| Hub vs switch?    | Hub repeats to all; switch learns MAC→port, delivers to one              |
| Switch vs router? | Switch = Layer 2, MAC, one LAN. Router = Layer 3, IP, between networks   |
| MAC vs IP?        | MAC = factory serial, local only. IP = assigned address, routed globally |
| OSI vs TCP/IP?    | OSI = 7 teaching layers. TCP/IP = 4 working layers                       |
| TCP vs UDP?       | TCP = certified mail, guaranteed. UDP = postcard, fast                   |

| Asked...                    | Answer in one breath                                                 |
|-----------------------------|----------------------------------------------------------------------|
| DNS / DHCP?                 | DNS = name→IP phonebook. DHCP = auto-addressing via DORA             |
| Port / socket?              | Port = door number. Socket = IP + port + protocol                    |
| NAT?                        | Many private devices share one public IP via the router's table      |
| Subnet mask?                | Marks network bits vs host bits; <code>/24</code> = 24 + 8           |
| Private ranges?             | <code>10/8</code> , <code>172.16/12</code> , <code>192.168/16</code> |
| Star topology?              | Every device to one switch — isolated failures, easy growth          |
| Fibre advantage?            | Light in glass: huge speed × distance, noise-immune                  |
| 3-way handshake?            | SYN → SYN-ACK → ACK, then data flows                                 |
| No internet — first checks? | Cable/Wi-Fi → got an IP? → ping the gateway?                         |

**You've got this. Tell the postal story, walk the website journey, and let the diagrams do the heavy lifting. Good luck tomorrow. 🍀**